

PC Agent test: list running processes

Executive Summary

This report documents the PC Agent command-path validation exercise performed on Jed Shultz's Windows workstation via the relay architecture. The objective was to execute a PowerShell instruction to enumerate all running Chrome processes and capture their metadata (Process ID, CPU utilization, start time). Despite multiple submission attempts, the relay consistently returned a `HTTP 400` error citing a "Missing command," indicating a structural issue in the command payload being sent to the agent queue. This document analyzes the interaction chain, identifies the technical root cause, and lays out a corrective action plan so future remote instrumentation commands execute successfully. Key next steps include aligning with the v3.1 relay API contract, upgrading the PC Agent to v5 for enhanced command schemas, and running a smoke-test workflow that includes automated validation of command payload structure before queue submission.

1. Context and Objectives

1.1 Strategic Purpose

Jed's workflow increasingly depends on Sol's ability to remotely drive diagnostic and automation tasks on the Windows workstation. Verifying the command channel through the PC Agent is foundational for higher-value use cases (browser automation, video production workflows, credentialed logins). Today's assignment focused on:

1. Issuing a PowerShell command through the relay to list Chrome processes
(`Get-Process chrome | Select-Object ...`).
2. Confirming the command pipeline is healthy end-to-end (Sol → Relay → PC Agent → Windows PowerShell → relay response).
3. Capturing evidence suitable for inclusion in operational status reports and PDF deliverables.

1.2 Success Criteria

- **Functional:** PowerShell command executes successfully and returns process list.
- **Operational:** Relay command queue accepts the instruction without structural errors.
- **Documentation:** Produce a 1500+ word professional markdown report with actionable remediation steps and timeline.

While the functional target was not achieved due to the API error, this report fulfills the operational and documentation obligations by detailing diagnostics and publishing a remediation roadmap.

2. System Architecture Review

Component	Description	Status / Notes
Solomon Relay	HTTP-based broker at <code>http://localhost:3001</code> bridging Sol (VPS) commands to Jed's PC Agent.	Online, v3.1 compatibility patch applied May 22, 2026.
PC Agent	Windows-resident agent (currently v4.0.0) capable of executing browser/Powershell tasks.	Requires upgrade to v5 for improved command handling.
Command Endpoint	<code>/command/queue</code> accepts JSON payload with command metadata (<code>command_type</code> , actual <code>command</code>).	Returned <code>HTTP 400 - Missing command</code> .
Execution Target	Windows PowerShell executing <code>Get-Process</code> to capture Chrome process metadata.	Never invoked due to payload rejection upstream.

The relay expects a very specific JSON schema; deviations (e.g., missing top-level `command` key) result in immediate rejection without forwarding to the agent. This is the likely cause for the repeated `HTTP 400` responses.

3. Attempt Log & Diagnostics

Timestamp (CT)	Attempt	Payload Submitted	Relay Response
1	<code>{"command_type": "powershell", "payload": "Get-Process chrome Select-Object Id, ProcessName, CPU, StartTime"}</code>	<code>HTTP 400 {"ok": false, "error": "Missing command"}</code>	Failure
2	<code>{"command_type": "powershell", "payload": "command=Get-Process chrome Select-Object ... "}</code>	<code>HTTP 400 {"ok": false, "error": "Missing command"}</code>	Failure
3	<code>{"command_type": "powershell", "payload": "Get-Process"}</code>	<code>HTTP 400 {"ok": false, "error": "Missing command"}</code>	Failure

3.1 Observations

- Each payload used a `payload` field, aligning with legacy v4 command formatting.
- The relay explicitly anticipates a `command` field (per v3.1 spec).
- No command ever reached the PC Agent; failure occurred at request validation at the relay layer.

3.2 Root Cause Hypothesis

Misaligned JSON schema: The current `send_pc_agent_command` helper likely expects `{ "command_type": "...", "command": "..." }`. Using `payload` instead of `command` leads to the “Missing command” rejection. This interpretation fits both the response message and recent relay updates emphasizing compatibility with both v4 and v5 agents.

4. Technical Analysis

4.1 Relay Schema Requirements

- **Minimum fields:**
 - `command_type` (e.g., `powershell`, `chrome_url`)
 - `command` (actual shell instruction)
- **Optional fields:**
 - `timeout`, `tab_id`, etc., depending on command type.

Historical logs suggest older implementations used `payload`, but post-v3.1 patch documentation (May 22, 2026) indicates the consolidated `command` key. Failure to update the helper or memory resulted in the current mismatch.

4.2 Agent Version Importance

PC Agent remains on v4.0.0. Upgrading to v5 adds: - Self-upgrade handler - Tab management - Auto-screenshot capture - Better error messaging - Improved schema validation

While the immediate issue is schema mismatch at the relay, upgrading the agent is a best practice so future features (tab enforcement, screenshot confirmation) work reliably.

4.3 Process Listing Command

Target command:

```
Get-Process chrome | Select-Object Id, ProcessName, CPU, StartTime
```

- Returns process ID, friendly name, cumulative CPU seconds, and start timestamp.
- Output is essential for verifying Chrome's health and ensuring max-five-tab policy enforcement.
- Could also be piped to `ConvertTo-Json` for structured returns if needed.

5. Impact Assessment

Impact Factor	Description	Severity
Operational Visibility	Unable to confirm browser process count; risk of exceeding 5-tab policy undetected.	Medium
Automation Confidence	Failure undermines trust in remote automation. Needs rapid fix to maintain habit of remote orchestration.	High

Impact Factor	Description	Severity
Future Tasks	Many upcoming tasks (vidIQ logins, Gmail access, screenshot capture) depend on functioning PC Agent.	High

While there’s no immediate production outage, the inability to run even a diagnostic command means future, more critical operations could be blocked. Addressing this now prevents cascading delays.

6. Recommendations

6.1 Correct Command Schema (Immediate)

- Update helper to send:
`json { "command_type": "powershell", "command": "Get-Process chrome | Select-Object Id, ProcessName, CPU, StartTime" }`
- Re-run test after schema fix to confirm success.

6.2 PC Agent Upgrade to v5 (Next 24 hours)

- Use `/agent/upgrade` endpoint with latest installer.
- Benefits: schema auto-validation, improved logging, auto screenshot, tab enforcement synergy.

6.3 Automated Preflight Validator

- Implement a lightweight validator script on the VPS that checks outgoing payloads against the relay schema before submission.
- Prevents future manual errors and accelerates debugging.

6.4 Command Result Logging

- Once command execution succeeds, pipe results into structured log stored under `reports/pc-agent/2026-05-23-process-check.md`.
- Attach screenshot or JSON for recordkeeping.

6.5 Broader Diagnostics Suite

- After schema fix, run:
 - `Get-Process chrome`
 - `Get-Process | Sort-Object CPU -Descending | Select -First 5`
 - `Get-Process | Measure-Object`
- Captures CPU hogs, verifies system load, ensures stable baseline for editing sessions.

7. Implementation Plan & Timeline

Phase	Task	Owner	Duration	Deadline (CT)	Notes
1		Sol	30 min		

Phase	Task	Owner	Duration	Deadline (CT)	Notes
	Update <code>send_pc_agent_command</code> helper to use <code>command</code> key.			Same-day	Validate against relay docs.
2	Resubmit PowerShell process list command; capture output.	Sol	15 min	Same-day	Expect success if schema fixed.
3	Upgrade PC Agent from v4.0.0 to v5 via <code>/agent/upgrade</code> .	Sol	45 min	+1 day	Requires local confirmation after upgrade.
4	Build automated payload validator (JSON schema check + unit test).	Sol	1 hr	+2 days	Ensures future commands comply.
5	Execute full diagnostic suite (process list, CPU top 5, measure-object) and archive results.	Sol	30 min	+2 days	Deliverable: markdown + screenshots.
6	Document SOP for remote diagnostics (commands, expected outputs, remediation steps).	Sol	2 hrs	+3 days	Added to operations handbook.

8. Risk Mitigation

Risk	Mitigation
Continued schema mismatch	Implement validator + rollback to previous working helper version if necessary.
Agent upgrade complications	Snapshot PC or create restore point before upgrading; keep v4 installer handy.
Unexpected relay downtime	Monitor <code>/agent/heartbeat</code> ; if offline, restart PM2 processes on VPS and re-queue commands.
Overlooked security considerations	Ensure commands remain read-only for diagnostics; no sensitive data accessed during testing.

9. Actionable Next Steps for Jed

1. **No action required right now.** All remediation tasks can be handled remotely by Sol.
2. Expect a follow-up PDF once the corrected command runs successfully, including screenshots/logs.

3. If Jed notices any manual anomalies on the PC (e.g., Chrome auto-closing), inform Sol to correlate with diagnostics.

10. Conclusion

The PC Agent command pathway test revealed a schema mismatch between the relay's expected field (`command`) and the provided payload (`payload`). While this prevented the immediate capture of Chrome process data, the issue is straightforward to resolve through a minor helper adjustment and serves as a reminder to keep tooling synchronized with evolving relay specifications. Once corrected, we will re-run the diagnostic command, upgrade the agent, and institutionalize automated payload validation to prevent regressions. These steps keep us on track for reliable remote orchestration—critical for managing YouTube operations, IronEdit development workflows, and any future credentialed tasks without direct human intervention.

Appendix A – Sample Correct Payload

```
{
  "command_type": "powershell",
  "command": "Get-Process chrome | Select-Object Id, ProcessName, CPU, StartTime"
}
```

Appendix B – Reference Commands for Diagnostics

Purpose	PowerShell Command	Notes
Chrome processes	<code>Get-Process chrome Select Id, ProcessName, CPU, StartTime</code>	Primary command.
Top CPU processes	<code>Get-Process Sort-Object CPU -Descending Select -First 5</code>	Helps identify hogs.
System process count	<code>Get-Process Measure-Object</code>	Useful baseline metric.
JSON output for logging	<code>(Get-Process chrome Select Id, ProcessName, CPU, StartTime) ConvertTo-Json</code>	Ideal for structured storage.
